

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2018

Subject: 341131 Electrical Drawing I

Section: 5-6

Date: 5 October 2018

Time: 13.00-15.00

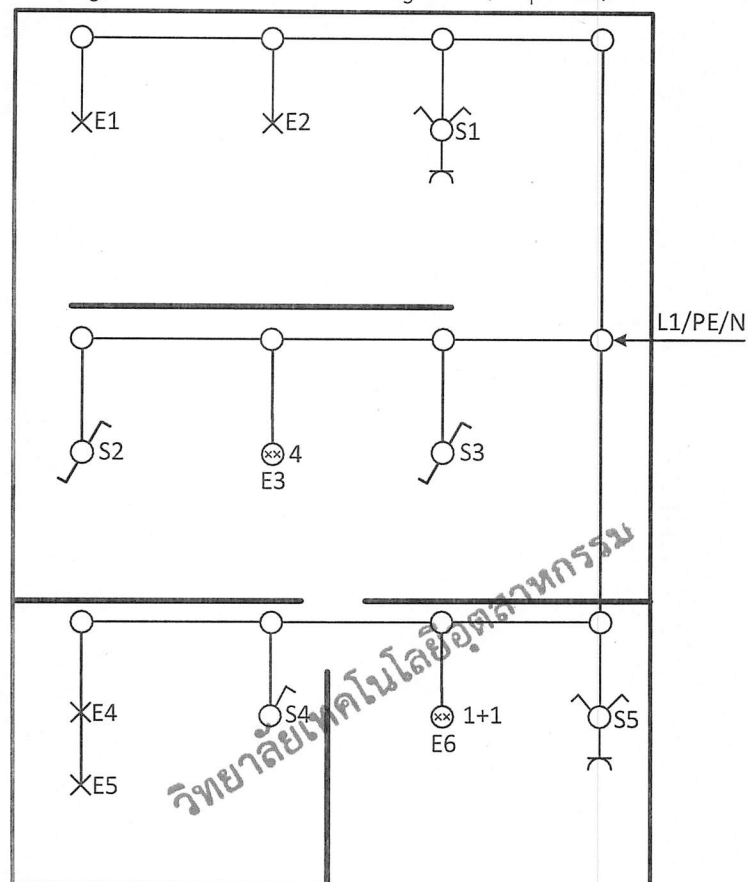
Name..... ID..... EP.....

Instructions:

1. The examination has 5 pages (including this page), 1 question and a total score of 40 points.
2. Write all your solution and answers on the answer sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculator is not allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.

Cheating in the exam is considered an extremely serious
offence which will result in expulsion from the University.

1. The floor plan with electrical circuit diagrams is shown below. It consists of 4 incandescent lamps (E1, E2, E4 and E5), 2 series incandescent lamps (E3 and E6), 2 series switches (S1 and S5), 1 switch (S4), 2 two-way switches (S2 and S3) and 2 plug sockets. Please draw the wiring, one-line and control diagrams. (40 points).



Remarks:

- E1 and E2 are controlled by S1.
- E3 is controlled by S2 and S3.
- E4 and E5 that are connected in parallel are controlled by S4.
- E6 is controlled by S5.
- 2 plug sockets are independent of each other.

Answer Sheet

One-line diagram (10 points) Name.....ID.....

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Wiring diagram (20 points) Name..... ID.....

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Control diagram (10 points) Name.....ID.....

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